

Project Initialization and Planning Phase

Date	05 July 2026
Team ID	
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI-powered loan approval prediction system using machine learning to provide fast and accurate loan decisions.
Scope	The system analyzes applicant details, predicts loan approval, and provides results through a Flask web application.
Problem Statement	
Description	Manual loan approval is time-consuming and may lead to inconsistent decisions.
Impact	An AI-based prediction system reduces processing time and improves decision accuracy.
Proposed Solution	
Approach	Train multiple machine learning models and deploy the best-performing XGBoost model using Flask.
Key Features	- Loan approval prediction,XGBoost machine learning model,Real-time prediction,User-friendly web interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel I5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Free
Software		
Frameworks	Backend	Flask
Libraries	Additional libraries	XGBoost, Scikit-learn, Pandas, NumPy
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Loan Prediction Dataset	Kaggle dataset, 614, csv UCI dataset, 690, csv